A Finite-Certificate Spectral Gap for the Square-Lattice Spin-$\frac{1}{2}$ Bilayer

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Abstract

We give an explicit finite-certificate proof of a uniform spectral gap for the antiferromagnetic spin-$\frac{1}{2}$ square-lattice bilayer. With the rung coupling normalized to one, every real intralayer coupling $0 \le g \le 1/10$ and every even periodic square of side $L \ge 256$ has a simple ground energy and spectral gap strictly greater than $\log(200/13)/1024$. The finite initialization uses cubic pair-creation trial states on 4×4, 8×4, and 8×8 rung tori. Exact signed residual identities, finite occupation-space comparisons, and rational polynomial certificates control the physical energy and partition function bounds uniformly throughout the coupling interval. A real symmetric spatial-transfer approximation supplies legitimate even-power moments without assuming nonnegative transfer spectra. Three staggered purities, six ordered unequal-error updates, and spatial interpolation then give the conclusion for all sufficiently large even squares, not only a dyadic sequence. The argument is formalized from the explicit Pauli Hamiltonian in Lean 4; its corrected source snapshot has been rebuilt and audited. No assertion of a near-critical parameter range, infinite-volume ground-state uniqueness, or global priority is made.

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1 Model, statement, and proof architecture

Let $\Lambda_{n,m} = (\mathbb{Z}/n\mathbb{Z}) \times (\mathbb{Z}/m\mathbb{Z})$. Throughout the physical argument, n, m are even and at least four. Let $\mathcal{E}_{n,m}$ contain the two positive-direction edges from each vertex; these are precisely the undirected nearest-neighbor edges counted once. There are $A = nm$ rungs, $2A$ spins, $2A$ edges, and Hilbert-space dimension 4^A. Put $S^\alpha = \sigma^\alpha/2$ for the usual Pauli matrices. We study

$$H_{n,m}(g) = \sum_{r \in \Lambda_{n,m}} \left(\boldsymbol{S}_{r,1} \cdot \boldsymbol{S}_{r,2} + \frac{3}{4}\mathbf{1} \right) + g \sum_{\{r,s\} \in \mathcal{E}_{n,m}} \sum_{a=1}^{2} \boldsymbol{S}_{r,a} \cdot \boldsymbol{S}_{s,a}. \tag{1}$$

The rung coupling is one. The additive constant changes no spectral difference. Write the eigenvalues, *including multiplicities*, as $E_0 \le E_1 \le \cdots \le E_{4^A-1}$. Write $Z_{n,m}(\beta) = \operatorname{Tr} e^{-\beta H_{n,m}}$ and $Q_{n,m}(\beta) = e^{\beta E_0} Z_{n,m}(\beta)$.

Theorem 1.1 (Uniform finite-volume gap)**.** *For every real $g \in [0, 1/10]$ and every even integer $L \ge 256$, the Hamiltonian (1) on $\Lambda_{L,L}$ has a simple lowest eigenvalue, and*

$$E_j - E_0 > \delta_* := \frac{\log(200/13)}{1024} \quad (1 \le j < 4^{L^2}). \tag{2}$$

Numerically, $\delta_* \approx 0.002669304696$; the logarithmic expression, not this decimal, is the proved constant.

The formal endpoint is `BilayerDressed.CubicPhysicalGap.standard_spin_uniform_gap`. It constructs a complete invertible complex eigenbasis for the explicit Pauli matrix and proves uniqueness and separation with multiplicities. There are no seed, table, residual, transfer-convergence, or ground-simplicity hypotheses remaining in this endpoint.

The proof has two independently intelligible parts. Sections 2–7 construct and certify three finite physical partition-function ratios. Sections 8–10 prove the dimension-independent propagation principle and apply it to those ratios. No large reference space is diagonalized numerically. Its role is an explicit finite-dimensional comparison in a proof.

Context and scope. Weak-coupling stability for product-state lattice systems is classical; see Yarotsky's Theorem 1 [3] and the higher-dimensional block-diagonalization result, Theorem 5.3, of Del Vecchio–Fröhlich–Pizzo–Rossi [2]. These results give important structural context, not the particular rational threshold certified here. The spatial-moment and coupled-purity strategy is motivated mathematically by the spin-one Haldane-gap manuscript [1], notably Proposition 3.1, Lemmas 4.1–4.2, Proposition 4.3, and its finite initializations in Section 5. The present proof constructs its own two-dimensional transfer approximants and all its model-specific finite inputs. It does not import the Haldane theorem as an axiom or reuse its numerical seed. The contribution established here is the displayed explicit interval and end-to-end finite certificate. We do not infer from this that no stronger published quantitative bound exists, nor that the physically expected entire dimer phase has been covered. Earlier narrower certificates for this same bilayer are proof-development comparisons, not a second physical model.

2 Exact rung representations

A rung is $\mathbb{C}^2 \otimes \mathbb{C}^2$. Its shifted exchange has eigenvalues $0, 1, 1, 1$. A convenient unnormalized basis matrix in the standard spin basis is

$$W = \begin{pmatrix} 0 & 0 & 1 & i \\ 1 & i & 0 & 0 \\ -1 & i & 0 & 0 \\ 0 & 0 & 1 & -i \end{pmatrix}, \qquad W^*W = 2\mathbf{1}, \qquad W^{-1} = \tfrac{1}{2}W^*.$$

Thus the first coordinate is the singlet and the other three coordinates are triplets. For the six single-spin components there are explicit rational real antisymmetric matrices G_a such that

$$S_a W = W(iG_a), \qquad W^{-1}(\boldsymbol{S}_1 \cdot \boldsymbol{S}_2 + 3/4)W = h := \operatorname{diag}(0, 1, 1, 1). \tag{3}$$

In particular the real lattice representative is

$$H^{\mathrm{R}} = \sum_r h_r - g \sum_{\{r,s\} \in \mathcal{E}} \sum_{a=1}^{6} G_{a,r} G_{a,s}. \tag{4}$$

Products at different sites are symmetric because $G_a^T = -G_a$. The six matrices and all sixteen local entries are fixed in `Gap.LocalRungBasis`; tensoring the exact inverse proves similarity to (1). Equivalently one may normalize W by $1/\sqrt{2}$ and obtain a unitary.

For the trial calculation use real hard-core creators $x_{r\alpha} = |\alpha\rangle_r\langle 0|$, $\alpha = 1, 2, 3$, and the triplet number $N = \sum_r n_r$, $n_r = \sum_\alpha |\alpha\rangle\langle\alpha|$. Creators at distinct sites commute; at the same site every product $x_{r\alpha}x_{r\gamma}$ is zero. They are not fermions. The Cartesian bond V_{rs} has the following complete action:

$$2V|00\rangle = \sum_\alpha |\alpha\alpha\rangle, \qquad 2V|\alpha 0\rangle = |0\alpha\rangle, \tag{5}$$

$$2V|0\alpha\rangle = |\alpha 0\rangle, \qquad 2V|\alpha\gamma\rangle = \delta_{\alpha\gamma}|00\rangle + |\gamma\alpha\rangle - \delta_{\alpha\gamma}\sum_\eta |\eta\eta\rangle. \tag{6}$$

Here and below $H^{\mathrm{C}} = N + g\sum V_{rs}$. The single globally defined diagonal involution $U(s) = (-1)^{\lfloor N(s)/2 \rfloor}$ conjugates H^{C} to H^{R}: it changes the sign of pair creation and annihilation, while preserving all number-conserving entries. This is a global basis change, not a product of inconsistent edgewise adapters. The vacuum $\Omega = |0 \cdots 0\rangle$ is fixed.

3 A finite spectral reference

3.1 Local comparison and occupation embedding

Put $P_{rs} = \sum_\alpha (x_{r\alpha}x_{s\alpha} + x_{r\alpha}^* x_{s\alpha}^*)$ and $V_{rs}^{\mathrm{n}} = V_{rs} - P_{rs}/2$.

Lemma 3.1 (Normal-bond lower bound)**.** $V_{rs}^{\mathrm{n}} \geq -(n_r + n_s)/2$. *Consequently, for* $g \geq 0$,

$$H^{\mathrm{C}} \geq (1 - 2g)N + \frac{g}{2} \sum_{\{r,s\} \in \mathcal{E}} P_{rs}. \tag{7}$$

Proof. For a real local vector with coefficients z_{ab}, twice the quadratic form of $V^{\mathrm{n}} + (n_r + n_s)/2$ is

$$\sum_{\alpha}(z_{\alpha 0} + z_{0\alpha})^2 + \sum_{\alpha<\gamma}(z_{\alpha\alpha} - z_{\gamma\gamma})^2 + \sum_{\alpha<\gamma}(z_{\alpha\gamma} + z_{\gamma\alpha})^2 + \sum_{\alpha\neq\gamma} z_{\alpha\gamma}^2.$$

This is nonnegative, and complex vectors follow by real and imaginary parts. Every rung belongs to four edges. Summing the local inequality therefore subtracts exactly $2gN$, proving (7). □

Let $I = \Lambda \times \{1, 2, 3\}$ be the $3A$ modes. Take the finite occupation space with basis $\nu \in \mathbb{N}^I$, $|\nu| \le A$. Its dimension is $\binom{4A}{A}$. Define the truncated creation matrix by $C_i|\nu\rangle = \sqrt{\nu_i + 1}|\nu + e_i\rangle$ if $|\nu| < A$, and zero otherwise. The isometry J_{hc} sends a physical configuration to the occupation vector with at most one particle, of its indicated color, at each rung. Compression reproduces the number and each distinct-site pair term exactly. Let D be the rung adjacency and $Z = (g/2)D \otimes I_3$. Then the right-hand side of (7) is the compression of

$$\mathcal{B} = (1 - 2g)N + \frac{1}{2}\sum_{ij} Z_{ij}(C_i C_j + C_j^* C_i^*). \tag{8}$$

An orthogonal mode change is legitimate at this finite cutoff. If $O^T O = I$, put $B_i = \sum_j O_{ji} C_j$ and define its columns by

$$U_O|\nu\rangle = (\nu!)^{-1/2} \prod_i B_i^{\nu_i}|0\rangle.$$

Finite normal ordering gives the inner product $\nu!\delta_{\nu\mu}$ before normalization. In that calculation annihilation crosses only shorter suffixes, so the top-degree boundary is never treated as a valid untruncated CCR. The columns are an orthonormal basis; U_O preserves degree and intertwines the creators, including their zero action on top-degree vectors. Choosing O as an actual orthogonal eigenbasis of Z diagonalizes (8) in its mode indices.

Include this total-degree space in the finite product space $\{0, \ldots, A\}^{3A}$, of dimension $(A + 1)^{3A}$. After the mode rotation, (8) is exactly the compression of a sum of one-mode matrices

$$h_M(\omega, z) = \omega N + \frac{z}{2}(a^{\dagger 2} + a^2), \quad M = A, \quad \omega = 1 - 2g. \tag{9}$$

This inclusion and compression are used only as finite linear maps; no infinite Fock representation is required.

3.2 Jacobi factorization and ordered eigenvalues

Lemma 3.2 (Finite Jacobi bound)**.** *For $b \ge 0$, $0 \le s \le t$, let $J_n(b, s, t)$ have diagonal $(s + t)k + sb$ and off-diagonal $-\sqrt{st(k + 1)(k + b)}$, $0 \le k < n$. Its increasing eigenvalues satisfy $\lambda_j \ge (t - s)j$ for $0 \le j < n$.*

Proof. Let A_n be upper bidiagonal with diagonal $\sqrt{s(k + b)}$ and upper diagonal $-\sqrt{t(k + 1)}$. Direct multiplication gives $A_n^* A_n = J_n$. The leading $(n - 1)$-dimensional principal block of $A_n A_n^*$ is $(t - s)I + J_{n-1}(b + 1, s, t)$. Only this block has the displayed form; the last row lacks an upper-diagonal contribution and must be removed. The square matrices $A_n A_n^*$ and $A_n^* A_n$ have the same characteristic polynomial, including zero multiplicities. Interlacing therefore gives $\lambda_{j+1}(J_n) \ge (t - s) + \lambda_j(J_{n-1}(b + 1, s, t))$. The lowest eigenvalue is nonnegative by the Gram factorization. Induction proves the claim, also when $s = 0$ or $b = 0$. □

Corollary 3.3 (Finite oscillator bound)**.** *If $|z| < \omega$, $\Omega = \sqrt{\omega^2 - z^2}$, and $e = (\Omega - \omega)/2$, then $\lambda_j(h_M(\omega, z)) \ge e + j\Omega$ for $0 \le j \le M$.*

Proof. In parity $p \in \{0,1\}$, write the occupation as $2k+p$. An appropriate diagonal sign change makes the off-diagonal entries negative. The block equals $J(b,s,t)+(e+p\Omega)I$, with $b=p+1/2$, $s=\omega-\Omega$, $t=\omega+\Omega$. Lemma 3.2 bounds its jth level by $e+(2j+p)\Omega$. Merging the two finite lists, with multiplicities, proves the assertion. □

Proposition 3.4 (Physical reference constants)**.** *On each ordinary seed torus, for $0 \le g \le 1/10$, set*

$$a = \frac{387}{500}, \qquad c_A(g) = -\frac{750A}{787}g^2. \tag{10}$$

There is an explicitly constructed finite isometry J such that $H^{\mathrm{R}} \ge J^*(c_A I + aN_{\mathrm{prod}})J$, where N_{prod} is the product occupation-label number. In particular $E_j \ge c_A + a$ for every $j \ge 1$. If $E_0 \le U$, then for $\beta > 0$,

$$Q_A(\beta) - 1 \le e^{\beta(U-c_A)}\left[(1-e^{-\beta a})^{-3A} - 1\right] \le 3Ae^{-\beta(a+c_A-U)}(1-e^{-\beta a})^{-3A-1}. \tag{11}$$

Proof. The eigenvalues z_i of Z satisfy $|z_i| \le 2g$ by its absolute row sum, and $\sum_i z_i^2 = \operatorname{Tr} Z^2 = 3Ag^2$ because $(D^2)_{rr} = 4$. Thus $\Omega_i \ge \sqrt{1-4g} \ge \sqrt{3/5} > a$ and

$$e_i = -\frac{z_i^2}{2(\omega+\Omega_i)} \ge -\frac{z_i^2}{2(\omega+a)}, \qquad \sum_i e_i \ge -\frac{750A}{787}g^2.$$

Tensoring the actual finite one-mode eigenbases gives the comparison. If Q is their product eigenbasis and I_A is the total-degree inclusion, the map from Cartesian states is $Q^T I_A U_O^T J_{\mathrm{hc}}$; compose it with the global involution to obtain the map from real rung states. The labels in N_{prod} are now ordered eigenvalue labels, not original site occupations. Compression min–max compares *ordered* physical levels with ordered reference levels. The number reference has a unique zero label. Removing that one label after excluding the physical lowest level yields the finite product tail; its geometric upper bound is the first inequality in (11). The second follows by integrating the derivative of $(1-x)^{-3A}$ on $[0,x]$. Neither ground vector is assumed to attain c_A. □

4 The cubic trial and its exact residual

On each seed torus let D denote its actual adjacency, including wraparound edges. Put $D_2 = \operatorname{off}(D^2) = D^2 - 4I$ and define

$$K = -\frac{g}{4}D + \frac{g^2}{8}(D_2 - D) - \frac{g^3}{64}(5D^3 - 4D_2 - 40D). \tag{12}$$

This symmetric matrix has zero diagonal. Define

$$S_K = \frac{1}{2}\sum_{r,s,\alpha} K_{rs}x_{r\alpha}x_{s\alpha}, \qquad \psi_K = e^{S_K}\Omega = \sum_{j=0}^{\lfloor A/2 \rfloor} \frac{S_K^j}{j!}\Omega. \tag{13}$$

The last equality is exact hard-core nilpotence. The vacuum coefficient is one, so $\|\psi_K\| \ge 1$. This is a specified polynomial trial state at every g, not a discarded-tail perturbation expansion.

For one edge $\{r,s\}$ put $x_\alpha = x_{r\alpha}$, $y_\alpha = x_{s\alpha}$, $k = K_{rs}$ and $u_\alpha = \sum_{t\neq r,s} K_{rt}x_{t\alpha}$, $v_\alpha = \sum_{t\neq r,s} K_{st}x_{t\alpha}$. All these variables commute; write $u\cdot v = \sum_\alpha u_\alpha v_\alpha$. Define the following *literal creation polynomials*:

$$q_2 = u\cdot v + \sum_\alpha x_\alpha(v_\alpha - 3ku_\alpha) + \sum_\alpha y_\alpha(u_\alpha - 3kv_\alpha) + (1-2k-3k^2)\sum_\alpha x_\alpha y_\alpha, \tag{14}$$

$$\begin{aligned} q_4 = & -\sum_\alpha x_\alpha u_\alpha(u\cdot v) - \sum_\alpha y_\alpha v_\alpha(u\cdot v) \\ & + \sum_{\alpha,\gamma} x_\alpha y_\gamma\left[v_\alpha u_\gamma - \delta_{\alpha\gamma}u\cdot v - u_\alpha u_\gamma - v_\alpha v_\gamma + 3ku_\alpha v_\gamma - k\delta_{\alpha\gamma}u\cdot v\right], \end{aligned} \tag{15}$$

$$q_6 = \sum_{\alpha,\gamma} x_\alpha y_\gamma u_\alpha v_\gamma (u\cdot v). \tag{16}$$

Each has its indicated creation degree; powers of g in its coefficients are not being truncated.

Proposition 4.1 (Exact transformed Hamiltonian)**.** *For any symmetric zero-diagonal K, define*

$$e_A = \frac{3g}{2}\sum_{\{r,s\}\in\mathcal{E}} K_{rs}, \quad R_2 = 2S_K + \frac{g}{2}\sum_{\mathcal{E}} q_2, \quad R_4 = \frac{g}{2}\sum_{\mathcal{E}} q_4, \quad R_6 = \frac{g}{2}\sum_{\mathcal{E}} q_6.$$

Then

$$(H^{\mathrm{C}} - e_A)\psi_K = e^{S_K}(R_2+R_4+R_6)\Omega = (R_2+R_4+R_6)\psi_K. \tag{17}$$

For (12), the three reference shifts are

$$\begin{aligned} e_{16} &= -12g^2 - 6g^3 - \tfrac{15}{2}g^4, \\ e_{32} &= -24g^2 - 12g^3 - \tfrac{45}{4}g^4, \\ e_{64} &= -48g^2 - 24g^3 - 15g^4. \end{aligned} \tag{18}$$

Proof. Split S_K into its outside-site part and the two-endpoint polynomial $\sum_\alpha(x_\alpha u_\alpha + y_\alpha v_\alpha + kx_\alpha y_\alpha)$. The outside part commutes with the bond. The exact local dressed vacuum has components 1, u_α, v_γ, and $u_\alpha v_\gamma + k\delta_{\alpha\gamma}$. Apply (5), then the inverse dressing obtained by negating u, v, k. Collecting the sixteen resulting entries gives $3k + q_2 + q_4 + q_6$ for twice the transformed bond. Also $e^{-S_K}Ne^{S_K}\Omega = 2S_K\Omega$. Summation proves the first identity. Every R_d is in the same commutative creation algebra as S_K, proving the second. Evaluating the edge entries of the explicitly given polynomial in D yields (18); the different fourth-order coefficients retain the actual short wrapping walks. □

The reference e_A is *not* the Rayleigh quotient. In particular, the nonzero degree-six term in (16) is necessary. For the cubic kernel the coefficient degrees reach g^7, g^{10}, and g^{13} in R_2, R_4, R_6, respectively, but the maximum creation degree remains six.

5 Finite norm and expectation estimates

5.1 Projection and finite graded norm bounds

Let $\widetilde{K} = K\otimes I_3$ and use the finite occupation creators to define $\mathcal{S}_B = \frac{1}{2}\sum_{ij} B_{ij}C_iC_j$ and $\Psi_B = \sum_{j=0}^{A/2}\mathcal{S}_B^j\Omega/j!$. The one-sided identity

$$J_{\mathrm{hc}}^* C_i = x_i J_{\mathrm{hc}}^*, \qquad \psi_K = J_{\mathrm{hc}}^*\Psi_{\widetilde{K}} \tag{19}$$

holds exactly. Creation cannot repair a configuration with two particles on a rung; hence the one-sided identity holds even on the orthogonal complement of the hard-core image. It does not assert multiplicativity of arbitrary compressions. For creation polynomials it gives the needed action intertwiner and a genuine Euclidean norm contraction.

Lemma 5.1 (Finite graded majorant)**.** *Suppose B is a real symmetric mode matrix, $B^2 \le qI$, and $\operatorname{Tr} B^2 \le T$, with $q, T \ge 0$. Define*

$$b_0 = 1, \qquad b_{j+1} = b_j \frac{qj + T/2}{j+1}, \qquad M_{A,d} = \sum_{j=0}^{A/2} \binom{2j+d}{d} b_j. \tag{20}$$

Then $\|\Psi_B\|^2 \le M_{A,0}$. For a homogeneous creation polynomial P_d, with faithful vacuum norm $\|P_d\Omega\|^2 \le F_d$, one has

$$\|P_d\Psi_B\|^2 \le F_d M_{A,d}. \tag{21}$$

Proof. On a safe grade the exact pair commutator is $[\mathcal{S}_B^*, \mathcal{S}_B] = d\Gamma(B^2) + \operatorname{Tr}(B^2)/2$. At the top two grades the missing creations contribute with the favorable nonpositive boundary correction. If A_{2j-2} bounds the squared creation norm on grade $2j-2$, finite adjoint duality gives $\|\mathcal{S}_B^*\xi_{2j}\|^2 \le A_{2j-2}\|\xi_{2j}\|^2$. The commutator therefore yields $A_{2j} \le A_{2j-2} + 2qj + T/2$, starting from $A_0 = T/2$. Summing this finite recurrence gives

$$\|\mathcal{S}_B\xi_{2j}\|^2 \le (j+1)(qj + T/2)\|\xi_{2j}\|^2.$$

Division by factorials yields (20); distinct grades are orthogonal. The finite symmetric-tensor multiplication estimate on grade n is $\|P_d\xi_n\|^2 \le \binom{n+d}{d}\|P_d\Omega\|^2\|\xi_n\|^2$. Here is an occupation-basis proof of that multiplication estimate. Write $r_\alpha = (P_d\Omega)_\alpha$ for $|\alpha| = d$, and let v_ν be the coefficients of ξ_n. For $|\gamma| = n + d$ the output coefficient is

$$(P_d\xi_n)_\gamma = \sum_{\substack{\alpha \le \gamma \\ |\alpha| = d}} r_\alpha v_{\gamma-\alpha} \sqrt{\frac{\gamma!}{\alpha!(\gamma-\alpha)!}}.$$

The multivariate Vandermonde identity gives $\sum_{\alpha\le\gamma, |\alpha|=d} \binom{\gamma}{\alpha} = \binom{n+d}{d}$. Cauchy–Schwarz therefore bounds the squared coefficient by $\binom{n+d}{d}\sum_\alpha |r_\alpha|^2|v_{\gamma-\alpha}|^2$. Summing over γ gives the product of the two squared norms; finite cutoffs only delete nonnegative output terms. Applying the estimate grade by grade and summing proves (21). The implemented occupation proof includes all cutoff-boundary cases; it never uses an untruncated CCR at top degree. □

For the cubic kernel the exact absolute row bound is $\kappa(g) = g + 2g^2 + 13g^3/4 \le 493/4000$. The squared row norms at the endpoint, which dominate the full interval, are

$$\rho_{16} = \frac{362417}{128000000}, \quad \rho_{32} = \frac{1443579}{512000000}, \quad \rho_{64} = \frac{143749}{51200000}. \tag{22}$$

The endpoint domination is algebraic: on these bipartite graphs,

$$|K| = \frac{g}{4}D + \frac{g^2}{8}(D_2 + D) + \frac{g^3}{64}(5D^3 + 4D_2 - 40D).$$

Every displayed coefficient matrix is nonnegative. On an edge $(D^3)_{rs} \ge 8$; on other entries the last assertion is immediate, and bipartite parity separates D_2 from D, D^3. Thus each absolute entry and squared row sum is monotone throughout the interval. Use $q = (493/4000)^2$ and $T_A = 3A\rho_A$. All sums in (20) are finite; the largest has 33 terms. They apply both to signed $\widetilde{K}$ and to its entrywise absolute value, without equating the two state norms.

5.2 Faithful residuals and the matching Gram matrix

For a distinct four-site set $S = \{s_0, s_1, s_2, s_3\}$, let Q_0, Q_1, Q_2 be the products of two color-contracted pair creators corresponding to 01|23, 02|13, 03|12. Collect the *signed* contributions of every bond and every repeated description of the same matching into $c_{S,m}(g)$. Then $R_4 = \sum_{S,m} c_{S,m} Q_m$. Terms with repeated sites vanish physically and are removed before choosing its faithful occupation lift.

Lemma 5.2 (Four-site Gram and translation count)**.** *For distinct sites,*

$$\langle Q_m\Omega, Q_{m'}\Omega\rangle = \begin{cases} 9 & m = m', \\ 3 & m \neq m', \end{cases} \qquad \|\sum_m c_m Q_m \Omega\|^2 = 6\sum_m c_m^2 + 3\left(\sum_m c_m\right)^2.$$

Different four-site supports are orthogonal. For any translation-invariant support weight $w(S)$ on an A-site torus,

$$\sum_{|S|=4} w(S) = \frac{A}{4} \sum_{\substack{|S|=4 \\ 0\in S}} w(S). \tag{23}$$

Proof. Each matching has nine color assignments. Two different matchings share exactly the three assignments in which all four colors agree. Distinct supports give distinct occupation supports. For (23), count pairs (r, S) with $r \in S$ in two ways and translate r to zero. This counts incidences, not translation orbits; it remains correct for supports with nontrivial stabilizers. □

An explicit local specification makes the signed coefficient collection independent of the table implementation. For site functions f, h define $P(f,h) = \sum_\alpha (\sum_i f_i x_{i\alpha})(\sum_j h_j x_{j\alpha})$. Extend the outside rows u, v by zero at both edge endpoints. For a matching m of a distinct support S, put

$$\mathcal{T}_m(f,h;j,l) = \sum_{\substack{(a,b,c,d) \text{ bijective onto } S \\ \{\{a,b\},\{c,d\}\}=m}} f_a h_b j_c l_d.$$

There are exactly eight ordered assignments. Expanding (15) gives

$$\begin{aligned} q_4 = &- P(e_r,u)P(u,v) - P(e_s,v)P(u,v) + P(e_r,v)P(e_s,u) \\ &- (1+k)P(e_r,e_s)P(u,v) - P(e_r,u)P(e_s,u) \\ &- P(e_r,v)P(e_s,v) + 3kP(e_r,u)P(e_s,v). \end{aligned}$$

The coefficient $c_{S,m}$ is exactly $g/2$ times the edge sum of these seven terms with each product $P(f,h)P(j,l)$ replaced by $\mathcal{T}_m(f,h;j,l)$. This explicitly defines the fiber sum before any absolute value.

The pair residual has a symmetric zero-diagonal coefficient F with $R_2 = \frac{1}{2}\sum_{r,s,\alpha} F_{rs} x_{r\alpha} x_{s\alpha}$ and $\|R_2\Omega\|^2 = (3A/2)\sum_s F_{0s}^2$. The sextic residual is lifted after deleting repeated-site terms. Define its unfiltered absolute scalar mass by

$$\ell_6(g) = \frac{g}{2} \sum_{\{r,s\}\in\mathcal{E}} \left(\sum_{t\neq r,s} |K_{rt}(g)|\right)^2 \left(\sum_{t\neq r,s} |K_{st}(g)|\right)^2.$$

Set $\kappa := 493/4000$. Each product of three color-contracted pairs on distinct sites has vacuum norm $\sqrt{27}$, giving $F_6 \le 27\ell_6(g)^2$. The entire outside sums are monotone in g, so they may first be moved to $g = 1/10$ and only then evaluated by subtracting the endpoint bond from κ. The endpoint

horizontal and vertical bond numerators over 64000 are $(1690, 1690)$, $(1685, 1690)$, and $(1685, 1685)$ on the three boxes. Hence the exact bound is

$$\ell_6(g) \le \frac{A}{20}\left[(\kappa - B_x/64000)^4 + (\kappa - B_y/64000)^4\right].$$

These are precisely the rational masses in Table 1 and its 64-rung companion in the source. This deliberately conservative bound requires no unjustified orthogonality between overlapping sextic descriptions.

5.3 A positive finite Wick upper bound

For the expectation estimate one must use the positive matrix $B = |K| \otimes I_3$, not a signed-kernel Gaussian identity. Let $\Psi_+ = \Psi_B$, $G = \|\Psi_+\|^2$, and

$$\tau_A = \frac{\kappa\rho_A}{1-\kappa^2}, \qquad L_{rs} = |K_{rs}| + \tau_A, \qquad \mathcal{L} = L \otimes I_3, \quad \kappa = 493/4000.$$

Lemma 5.3 (Finite Wick domination)**.** *For every even list of mode indices $I = (i_1, \dots, i_{2d})$,*

$$\langle \Psi_+, C_{i_1} \cdots C_{i_{2d}} \Psi_+ \rangle \le G \sum_{\pi} \prod_{\{a,b\}\in\pi} \mathcal{L}_{i_a i_b}, \tag{24}$$

where π runs through the perfect matchings of the list positions.

Proof. All occupation coefficients are nonnegative. The exact truncated commutator satisfies $C_i^* C_j \le C_j C_i^* + \delta_{ij} I$ entrywise. Expanding the finite Taylor polynomial gives $C_i^* \Psi_+ \le \sum_j B_{ij} C_j \Psi_+$. Moving one annihilator through a creation word and then using this inequality again yields, for an odd list I, the actual recurrence

$$f_i \le \sum_{k\in I} B_{i,i_k} F_{I\setminus k} + \sum_j (B^2)_{ij} f_j, \quad f_i = \langle \Psi_+, C_i C_I \Psi_+ \rangle.$$

The sum is over occurrences, so repeated indices are counted correctly. For the rung block, symmetry and the squared-row bound imply $(|K|^2)_{ij} \le \rho_A$ and $(|K|^3)_{ij} \le \kappa\rho_A$. Consequently $L \ge |K| + |K|^2 L$ entrywise and the row sums of $|K|^2$ are at most $\kappa^2 < 1$. If a finite vector satisfies $h \le B^2 h$, its positive maximum would satisfy $h_i \le \kappa^2 h_i < h_i$; hence $h \le 0$. Induction on the word length, comparing f with the perfect-matching expression, now proves (24). The empty-word moment is exactly G. This is an upper bound for a finite Taylor state, not an exact Gaussian Wick identity. Color blocks remain separate throughout. □

Coefficientwise absolute values and (19) give $|\psi_K| \le J_{\mathrm{hc}}^* \Psi_+$. Replacing a signed creation polynomial by its coefficientwise absolute majorant and deleting the hard-core projection can only increase the resulting nonnegative scalar product. Thus Lemma 5.3 bounds the *actual physical signed residual mean*. For four sites put $p_m = \prod_{\{a,b\}\in m} L_{s_a s_b}$. Summing the two free colors gives the weight

$$9p_m + 3\sum_{m'\neq m} p_{m'} = 6p_m + 3\sum_{m'} p_{m'}. \tag{25}$$

For six sites, 27 color assignments and 15 positional matchings give the upper bound $405 \max_{rs} L_{rs}^3$ per unit scalar mass. All these mean estimates retain the multiplier $G \le M_{A,0}$. No equality of the signed and positive state norms is asserted.

6 Exact finite graph certificates

The preceding lemmas reduce the required physical bounds to finite rational polynomial arithmetic, but this reduction alone is not a seed. We describe how the stored certificates are tied to the actual graphs.

On a torus, enumerate sites row-major, with 0 fixed. The cubic entries in (12) have integer coefficient arrays after scaling by 64. Expanding (14)–(16) produces exact rational polynomials. For the quartic term, every support containing 0 is present, including those whose coefficient vanishes. There are respectively $\binom{15}{3} = 455$, $\binom{31}{3} = 4495$, and $\binom{63}{3} = 39711$ supports. Each has three matching labels. Signed fiber sums are collected *before* absolute values or Gram weights are applied. Their degree-at-most-ten polynomials are represented by eleven integer coefficients with common normalization 524288. The three displayed envelope dictionaries have 43, 508, and 462 entries; these are coefficient classes, not counts of physical supports or numerical samples.

Proposition 6.1 (Whole-interval certification)**.** *Every stored coefficient bound and residual norm bound used below holds for all real* $0 \le g \le 1/10$*. The certificate hypotheses are discharged by exact finite graph identities and rational inequalities.*

Certificate argument. For each coefficient dictionary entry $c(g)$, the proof supplies a rational u and checks positivity of $u - c(g)$ and $u + c(g)$ on the entire interval. After the change $x = 10g$, represent a polynomial as

$$p(x) = \sum_{k=0}^{d} b_k \binom{d}{k} x^k (1-x)^{d-k}.$$

Nonnegative rational Bernstein coefficients imply $p(x) \ge 0$ for $0 \le x \le 1$. The basis-conversion identity and coefficient inequalities are proved exactly. The pair squared-norm polynomials and the collected signed quartic Gram polynomials are treated in the same way; the latter are not replaced by the sum of absolute local terms.

Separately, exact table theorems identify each actual graph coefficient with its dictionary polynomial, up to its stored sign. Row validity, increasing codes, and exact binomial cardinality prove that the rooted support enumeration is a bijection. Bounded chunks and balanced concatenation retain every row and every boundary comparison. They change evaluation cost, not the mathematical condition. The covariance cache is proved equal to $|K_{rs}(1/10)| + \tau_A$ for every pair by actual displacement translation. The entrywise monotonicity proved above bounds $L(g)$ by this endpoint covariance on the whole interval. When a translated support is reordered, the same explicit permutation transports its matching label, coefficient, and covariance weight. Therefore (23) applies to both the Gram and weighted-mean sums without assuming enumeration invariance. Exact integer aggregates then prove the rational ceilings.

The generated arrays are machine-readable appendices in the formal source. Ordinary Python generation and diagnostic checks help produce these witnesses; neither their exit status nor an imported JSON number is an axiom. The accepted Lean terms prove graph membership, interval inequalities, and aggregate values. □

For clarity, the quantities $F_{A,d}$ in Table 1 bound *squared faithful vacuum norms*. They are multiplied by $M_{A,d}$ before square roots are taken. The actual physical component bounds follow from (19) and (21). Only areas 16 and 32 need residual norms; area 64 needs its mean but no Temple variance estimate.

	$A = 16$	$A = 32$
$F_{A,2}$	$\frac{2871661602999}{819200000000000000}$	$\frac{10257117031323}{2621440000000000000}$
$F_{A,4}$	$\frac{10086255523899002668131}{419430400000000000000000000}$	$\frac{61781177216998702643223}{1342177280000000000000000000}$
$\ell_{A,6}$	$\frac{92232993647601}{655360000000000000}$	$\frac{2956223510791297}{10485760000000000000}$
$F_{A,6}$	$27\ell_{16,6}^2$	$27\ell_{32,6}^2$

Table 1: Exact finite vacuum certificates.

Proposition 6.2 (Actual cubic trial bounds)**.** *Let* $\mu_A = \langle\psi_K, H^{\mathrm{C}}\psi_K\rangle/\|\psi_K\|^2$. *Uniformly on* $[0, 1/10]$*, the actual unnormalized bounds*

$$|\langle\psi_K, (H^{\mathrm{C}} - e_A)\psi_K\rangle| \le m_A, \qquad \|(H^{\mathrm{C}} - e_A)\psi_K\| \le \varepsilon_A$$

hold with the rational ceilings in Table 2. Hence $|\mu_A - e_A| \le m_A$*, and the normalized variance is at most* ε_A^2 *where a norm bound is supplied.*

Proof. Use the exact residual identity, the faithful norm and positive mean estimates, and the whole-interval graph certificates. For $d = 2, 4, 6$ choose rational $\eta_{A,d} > 0$ with $F_{A,d}M_{A,d} \le \eta_{A,d}^2$; exact squaring checks give $\sum_d \eta_{16,d} < 10982/10^6$ and $\sum_d \eta_{32,d} < 20172/10^6$. The finite Wick coefficient aggregates multiplied by the separately bounded positive norm $M_{A,0}$ give m_A. Finally $\|\psi_K\| \ge 1$ and centering at the actual Rayleigh quotient cannot increase the normalized residual variance. This last step does not identify e_A with μ_A. □

Box	$M_{A,0}$ upper bound	m_A	ε_A
4×4	$1070873423/10^9$	$167/10^6$	$10982/10^6$
8×4	$286527607/250000000$	$346/10^6$	$20172/10^6$
8×8	$328014697/250000000$	$800/10^6$	not required

Table 2: Uniform actual trial ceilings. Norms are Euclidean Hilbert norms.

7 Energy intervals and the physical seed

7.1 Temple's inequality with an absolute threshold

For a normalized vector ϕ, let $\mu = \langle\phi, H\phi\rangle$ and $\sigma^2 = \|(H - \mu)\phi\|^2$. If every nonlowest ordered eigenvalue is at least $\theta > \mu$, the finite-dimensional Temple inequality is

$$E_0 \ge \mu - \frac{\sigma^2}{\theta - \mu}. \tag{26}$$

Indeed, $(E - E_0)(E - \theta) \ge 0$ at every eigenvalue, including the lowest one. Taking its expectation gives $\sigma^2 \ge (\mu - E_0)(\theta - \mu)$. The input θ is an *absolute energy threshold*, not a presumed gap. Proposition 3.4 supplies $\theta = c_A + a$ independently of the trial. The following positive rational numbers satisfy, throughout $[0, 1/10]$, $d_A \le c_A + a - e_A - m_A$:

$$d_{16} = \frac{588758821}{787000000}, \qquad d_{32} = \frac{568075073}{787000000}, \qquad d_{64} = \frac{5263369}{7870000}. \tag{27}$$

For example, subtracting the endpoint value of $c_A + a - e_A - m_A$ leaves $(1/10 - g)$ times an explicitly nonnegative polynomial; `ScalarInterval.denominator16_lower` and its two companions prove this whole-interval statement. There is no endpoint-sampling inference. Combining (26) with Proposition 6.2 gives

$$e_A - m_A - \varepsilon_A^2/d_A \le E_A \le e_A + m_A \quad (A = 16, 32), \qquad E_{64} \le e_{64} + m_{64}. \tag{28}$$

Here E_A denotes the ground energy of its specified box, not an energy assumed to depend only on area.

7.2 Separate spatial defects and thermal tails

Define the two finite ground-energy defects

$$d_x^{\mathrm{phys}} = E_{8,4} - 2E_{4,4}, \qquad d_y^{\mathrm{phys}} = E_{8,8} - 2E_{8,4}.$$

The extensive quadratic and cubic terms cancel exactly in (18): $e_{32} - 2e_{16} = 15g^4/4$ and $e_{64} - 2e_{32} = 15g^4/2$. Using (28) yields the fixed bounds

$$\begin{aligned} d_x^{\mathrm{phys}} &\le \overline{d}_x = \frac{810971762131}{588758821000000} = \frac{3}{8000} + m_{32} + 2m_{16} + 2\varepsilon_{16}^2/d_{16}, \\ d_y^{\mathrm{phys}} &\le \overline{d}_y = \frac{957049999441}{284037536500000} = \frac{3}{4000} + m_{64} + 2m_{32} + 2\varepsilon_{32}^2/d_{32}. \end{aligned} \tag{29}$$

This correlated cancellation is why separate extensive estimates are not sufficient: their errors would survive the subtraction. Keeping two different defects also avoids an unnecessarily severe common-error seed.

At $\beta = 16$, apply (11) with the actual variational upper $U_A = e_A + m_A$. Exact positive Taylor sums through degree 50 prove

$$e^{16a} > 238947, \quad e^{16d_{16}} > 157894, \quad e^{16d_{32}} > 103691, \quad e^{16d_{64}} > 44383.$$

Substitution in the last expression of (11) gives

$$Q_{4,4} < \overline{Q}_0 = \frac{200061}{200000}, \quad Q_{8,4} < \overline{Q}_1 = \frac{1000927}{1000000}, \quad Q_{8,8} < \overline{Q}_2 = \frac{1004331}{1000000}. \tag{30}$$

The inequalities for the Taylor sums, their application to the real exponential, and the subsequent rational powers are separate proved steps. Floating-point exponential evaluation is not used for acceptance.

For a volume $v = (n, m, \beta)$ define the staircase ratios

$$A(v) = \frac{Z_{2n,m}(\beta)}{Z_{n,m}(\beta)^2}, \quad B(v) = \frac{Z_{2n,2m}(\beta)}{Z_{2n,m}(\beta)^2}, \quad C(v) = \frac{Z_{2n,2m}(2\beta)}{Z_{2n,2m}(\beta)^2}. \tag{31}$$

These ratios are not yet assumed to be purities; that is established in Section 8. Direct cancellation of the ground-energy factors and $Q \ge 1$ give at $v_0 = (4, 4, 16)$

$$A(v_0) \ge \frac{e^{-16\overline{d}_x}}{\overline{Q}_0^2}, \qquad B(v_0) \ge \frac{e^{-16\overline{d}_y}}{\overline{Q}_1^2}, \qquad C(v_0) \ge \frac{1}{\overline{Q}_2^2}.$$

The lower Taylor polynomial $e^{-z} \ge 1 - z + z^2/2 - z^3/6$ for $z \ge 0$ now provides an entirely rational initialization.

Proposition 7.1 (Actual initial seed)**.** *For every $g \in [0, 1/10]$,*

$$1 - A(v_0) \le \frac{11197089}{500000000}, \quad 1 - B(v_0) \le \frac{27119173}{500000000}, \quad 1 - C(v_0) \le \frac{8606051}{1000000000}. \tag{32}$$

All five finite inputs needed for (28), and therefore all three inputs in (32), are proved for the actual Hamiltonian. The statement is not conditional on a numerical oracle. At $g = 0$ the checks have a useful exact consistency limit: $Z_{n,m}(\beta) = (1 + 3e^{-\beta})^{nm}$, $A = B = 1$, and $C = [(1 + 3e^{-2\beta})/(1 + 3e^{-\beta})^2]^{4nm}$. The true gap is then one; the present uniform constant is intentionally much less sharp.

8 Spatial moments without a positive transfer spectrum

Fix $\beta > 0$, $g \ge 0$, and an even transverse width $m \ge 4$. Group the real representative (4) into columns. The column Hilbert space is finite; let h_m contain its rung and vertical interactions, and let G_b be the real insertions on its sites. Then the strip Hamiltonian has the form $\sum_j h_{m,j} - g \sum_{j,b} G_{b,j} G_{b,j+1}$. The width dependence of h_m, the insertions, and the transfer dimension is allowed. No bounded-width approximation is taken.

For $q \ge 1$ time slices put $\epsilon = \beta/q$, and let an event label be either $\varnothing$ or one insertion label b. Define the symmetric two-history vertex matrix

$$W_\epsilon(x, y) = \begin{cases} I - \epsilon h_m, & x = y = \varnothing, \\ \sqrt{\epsilon g}\, G_b, & \{x, y\} = \{\varnothing, b\}, \\ 0, & x \ne \varnothing,\ y \ne \varnothing. \end{cases} \tag{33}$$

For histories $\xi, \eta \in (\{\varnothing\} \cup \mathcal{B})^q$, define

$$(X_q)_{\xi\eta} = \sum_{a_0, \dots, a_{q-1}} \prod_{t=0}^{q-1} W_\epsilon(\xi_t, \eta_t)_{a_t, a_{t+1}}, \qquad a_q = a_0. \tag{34}$$

Proposition 8.1 (Constructed finite spatial moment limits)**.** *Each X_q is real symmetric. For each fixed even physical circumference $n \ge 4$,*

$$\lim_{q \to \infty} \operatorname{Tr} X_q^n = Z_{n,m}(\beta). \tag{35}$$

The same statement holds with the two spatial directions interchanged.

Proof. The symmetry of (33) in its two event labels proves $(X_q)_{\xi\eta} = (X_q)_{\eta\xi}$ directly. Neither entrywise positivity nor transposition of G_b is used. Contracting the same finite rectangular tensor network first in time or first in space gives $\operatorname{Tr} X_q^n = \operatorname{Tr} P_n(\epsilon)^q$. Here P_n is a finite matching polynomial: occupied bonds at a time slice cannot share a vertex, and unoccupied vertices supply $I - \epsilon h_m$. Its constant term is I and its first derivative is $-H_{n,m}$. Every selected edge has two square-root factors whose product is ϵg. Thus $P_n(t) = I - tH_{n,m} + O(t^2)$ in a finite matrix algebra. A noncommutative telescoping estimate for powers implies $P_n(\beta/q)^q \to e^{-\beta H_{n,m}}$ in norm. Trace continuity proves (35). The column-to-torus reindexing is an exact equality of onsite and edge lifts. Swapping the two spatial axes gives the second direction. □

The formal route uses these finite symmetric approximants and their trace limits. It does not need to construct a single infinite-dimensional limiting transfer operator or prove its spectral

completeness. For even n, if $\lambda_{q,i}$ are the real eigenvalues of X_q and the denominator is nonzero, the weights

$$p_{q,i} = \frac{\lambda_{q,i}^n}{\sum_j \lambda_{q,j}^n} \tag{36}$$

are nonnegative and sum to one, even if some $\lambda_{q,i} < 0$. The denominators are positive eventually, since their limits are the strictly positive physical partition functions. Consequently $\operatorname{Tr} X_q^{2n}/(\operatorname{Tr} X_q^n)^2$ is a genuine probability purity. Passing the finite inequalities to the limit legitimizes the spatial ratios in (31). Thermal purity is already the purity of the finite Gibbs weights $e^{-\beta E_j}/Z(\beta)$.

This argument isolates the two-dimensional issue: each fixed transverse width has its own transfer spectrum, but the probability inequalities used below have constants independent of its dimension or width. No common eigenbasis, joint x/y spectrum, or algebraic relation between different widths is assumed.

9 Three-coordinate propagation

For any positive partition function on $v = (n, m, \beta)$, let d_i double coordinate $i \in \{x, y, t\}$ and set $P_i(v) = Z(d_i v)/Z(v)^2$. For the present model, Proposition 8.1 and the Gibbs representation prove $0 < P_i \le 1$ at all even spatial sizes at least four.

Lemma 9.1 (Purity squaring and cross transport)**.** *For a nonnegative finite probability vector p with purity $P = \sum_i p_i^2$,*

$$1 - \sum_i (p_i^2/P)^2 \le \frac{(1-P)^2}{2P^2}.$$

For every pair of coordinates,

$$P_i(v)^2 P_j(d_i v) = P_j(v)^2 P_i(d_j v), \qquad P_i(d_j v) \ge P_i(v)^2 P_j(d_i v). \tag{37}$$

Proof. Expansion of nonnegative finite sums gives $2[(\sum p_i^2)^2 - \sum p_i^4] \le [(\sum p_i)^2 - \sum p_i^2]^2$. Divide by $2P^2$. Equation (37) follows by cancellation of the four partition functions in a commuting coordinate square; its inequality uses $P_j(v) \le 1$. Both statements pass to the spatial moment limits. □

Let $a = 1 - A$, $b = 1 - B$, $c = 1 - C$ for the staircase (31). Define

$$u(x, y, z) = 1 - (1-x)^4 (1-y)^2 (1-z), \qquad F(x, y, z) = \frac{u(x, y, z)^2}{2(1 - u(x, y, z))^2}. \tag{38}$$

For clarity, the first two cross transports are

$$P_x(d_y v) \ge A^2 B, \qquad P_x(d_t d_y v) \ge A^4 B^2 C.$$

The second inequality uses the shifted staircase anchors for B and C. Apply own-x purity squaring at $d_t d_y v$; then relabel the coordinates cyclically at the subsequent shifted anchors. Lemma 9.1 gives at $v' = (2n, 2m, 2\beta)$ the *ordered* inequalities

$$a' \le F(a, b, c), \qquad b' \le F(b, c, a'), \qquad c' \le F(c, a', b'). \tag{39}$$

In the second and third expressions the new preceding bounds must be used. This is not a simultaneous update of three old errors.

Proposition 9.2 (Six exact unequal-error warmups)**.** *Starting from Proposition 7.1, six applications of (39) give the rational upper errors in Table 3. At $(n, m, \beta) = (256, 256, 1024)$ every error is less than $1/400$.*

Proof. The function F is increasing in each argument on $[0, 1)^3$. Each table entry is an upward rational enclosure of the corresponding update, proved by exact arithmetic. The coordinates double six times. □

k	$10^9 a_k$	$10^9 b_k$	$10^9 c_k$
0	22394178	54238346	8606051
1	27524545	47333988	11099297
2	30210971	39236906	12360104
3	28826046	28539483	10813383
4	20840176	14774239	5588084
5	8072381	3365491	919811
6	836295	132733	15137

Table 3: Exact rational enclosures. All entries denote rational numbers, not rounded measurements.

For a common error ceiling u, the update obeys

$$F(u, u, u) = G_7(u)u^2, \qquad G_7(u) = \frac{1}{2}\left[\sum_{k=1}^{7}(1-u)^{-k}\right]^2.$$

The exact checks give $G_7(1/400) < 26$. Set $K_* = 26$ and $r_* = 13/200$, so $r_*/K_* = 1/400$ and $r_* < 1$. Induction, including the ordered nature of (39), gives at scale j after the warmup

$$1 - A,\ 1 - B,\ 1 - C \le u_j := \frac{r_*^{2^j}}{K_*}, \quad (n, m, \beta) = (256 \cdot 2^j, 256 \cdot 2^j, 1024 \cdot 2^j). \tag{40}$$

The fact $K_* u_j < 1$ ensures each new component remains within the old common ceiling while the other components are being updated.

10 All-even-size interpolation and the gap

Lemma 10.1 (Rectangular interpolation)**.** *Let $n \le L \le 2n$ and $m \le M \le 2m$, all four sizes even and at least four. Then*

$$P_t(L, M, \beta) \ge [A(n, m, \beta)^4 B(n, m, \beta)^2 C(n, m, \beta)]^{LM/(4nm)}. \tag{41}$$

Proof. For a finite real symmetric transfer matrix, monotonicity of finite ℓ^p norms gives, for $n \le L$, the upper moment bound $\operatorname{Tr} X^L \le (\operatorname{Tr} X^n)^{L/n}$ at even exponents; reversing the endpoint choice gives the corresponding lower bound. Pass to the positive moment limits and apply the estimates in each spatial direction. At temperature β,

$$Z_{L,M}(\beta) \le Z_{n,m}(\beta)^{LM/(nm)}.$$

At temperature 2β, use the upper endpoint to obtain $Z_{L,M}(2\beta) \ge Z_{2n,2m}(2\beta)^{LM/(4nm)}$. Finally $A^4 B^2 C = Z_{2n,2m}(2\beta)/Z_{n,m}(\beta)^8$ by exact cancellation. Dividing proves (41). □

Proof of Theorem 1.1. For an arbitrary even $L \geq 256$, choose the unique dyadic size box containing it: $n_j = 256 \cdot 2^j \leq L < 2n_j$. The exponent in (41) is at most one, and $0 < A, B, C \leq 1$. Therefore (40), Bernoulli's inequality, and Lemma 10.1 show that the actual finite Gibbs purity at $\beta_j = 1024 \cdot 2^j$ satisfies

$$P_t(L, L, \beta_j) \geq (1 - u_j)^7 \geq 1 - 7u_j. \tag{42}$$

Let w_0 be a Gibbs weight at a lowest energy. Since $\sum w_i^2 \leq \max_i w_i$, $w_0 \geq 1 - 7u_j$. For any other index k, including a hypothetically degenerate lowest level,

$$e^{-\beta_j(E_k - E_0)} = \frac{w_k}{w_0} \leq \frac{7u_j}{1 - 7u_j} < r_*^{2^j}.$$

The last strict inequality follows from the exact margin $7 < K_*(1 - 7r_*/K_*)$. Taking logarithms and dividing by $1024 \cdot 2^j$ gives (2). A second lowest eigenvector would contradict its strictly positive right-hand side, proving simplicity. The exact rung similarity then transports the complete spectrum and multiplicities to the Pauli Hamiltonian. □

There is no exchange of the thermodynamic and zero-temperature limits in this argument. For each finite L, a finite synchronized temperature is selected from its spatial size box. This proves the uniform finite-volume statement without asserting a unique infinite-volume GNS ground state. Odd lengths, open boundaries, the near-critical regime, and a formula for the full spectrum are outside the theorem.

A Additional exact certificate data

For $q = (493/4000)^2$, the finite recurrence (20) uses

$$T_{16} = \frac{1087251}{8000000}, \quad T_{32} = \frac{4330737}{16000000}, \quad T_{64} = \frac{431247}{800000}.$$

Convenient upper bounds for the remaining norm moments are

A	$M_{A,2}$	$M_{A,4}$	$M_{A,6}$
16	726404409/500000000	1118930283/500000000	1794975549/500000000
32	$1981768181/10^9$	$387485587/10^8$	$7533402531/10^9$
64	828790239/250000000	544139463/62500000	169920309/8000000

The actual kernel check gives both $qI - \widetilde{K}^2 \geq 0$ and its positive absolute-kernel counterpart, with trace bounds T_A. The endpoint covariance constants are (22) inserted into τ_A, not estimates of unknown infinite-volume correlations.

The pair coefficient can also be generated by a compact matrix expression. Let $Z_0 = \operatorname{diag}((DK)_{rr})$ and use $\circ$ for entrywise multiplication. The symmetric coefficient F in Section 5 is exactly

$$F = 2K + \frac{g}{2}D + \frac{g}{2}\operatorname{off}(DK + KD) - g(D \circ K) + \frac{g}{2}\operatorname{off}(KDK) - 2g(Z_0K + KZ_0) + 2g(D \circ K \circ K). \tag{43}$$

The finite regrouping from the literal edge polynomial (14) to (43) is proved separately. It is not substituted as an ansatz. For translation-invariant K, its rooted row then gives the exact pair vacuum norm and mean coefficient used by the table.

The quartic mean integer aggregate is, explicitly,

$$\frac{A}{4} \sum_{\substack{S \ni 0 \\ |S|=4}} \sum_{m=0}^{2} u_{S,m} \left(6p_{S,m} + 3 \sum_{m'=0}^{2} p_{S,m'} \right),$$

where $u_{S,m}$ is the proved whole-interval envelope of the *signed* canonical coefficient and every p uses the exact endpoint covariance. In the integer implementation the denominator is the envelope denominator multiplied by the square of the covariance denominator. The normalization 524288 has already been included in the coefficient envelope and is not applied a second time. The sextic aggregate is bounded by $405\ell_{A,6}(\max L)^3$. The final physical mean is the sum of the pair, quartic and sextic aggregates times $M_{A,0}$.

B Proof and certificate interfaces

The repository separates mathematical source, finite certificate witnesses, and reproduction evidence. The abbreviated correspondence below is expanded, with exact theorem names, in the companion proof map.

Mathematical step	Formal source family
Pauli/rung similarity	`Gap.LocalRungBasis`, `Gap.PhysicalGapTheorem`
Finite reference and modes	`Dressed.CoupledQuadraticComparison`, `Dressed.FiniteOccupationModeRotation`, `Dressed.PhysicalProductComparison`
Jacobi ordered levels	`Dressed.FiniteJacobi`, `Dressed.FiniteOscillatorSpectrum`
Exact transformed residual	`Dressed.GlobalResidualDegrees`, `Dressed.GlobalDressedResidual`, `Dressed.CreationResidualAction`
Finite graded/Wick bounds	`Dressed.FinitePairNormRecursion`, `Dressed.FinitePositiveMoments`, `Dressed.FiniteWickPairings`
Actual signed graph certificates	`Dressed.CubicResidualIntervalBounds`, `Dressed.QuarticIntegerActualNorm16/32`, `Dressed.QuarticIntegerActualMean16/32/64`
Energy and thermal seed	`Dressed.CubicTrialSeedCriterion`, `Dressed.CubicSeedThermal`
Constructed spatial moments	`Gap.HistoryTransfer`, `Gap.HistoryApproximation`, `Gap.HistoryStripTorus`
Six ordered warmups	`Dressed.CoordinateWarmup`, `Dressed.ScalarWarmupCertificate`
All-even-size conclusion	`Gap.LimitCriterion`, `Dressed.CubicPhysicalGap`

C Verification, limitations, and reproducibility

The formal proof follows the mathematical dependencies above: explicit Pauli operators, finite rung and occupation maps, local residual algebra, exact finite coefficient certificates, physical seed ratios, and the all-size purity criterion. Formalization is not a substitute for checking that these definitions represent the intended model; the explicit basis maps and edge lists provide that semantic connection. The companion `docs/proof-map.md` records specific theorem names and source locations. Machine-readable polynomial arrays are appendices to the finite certificate, rather than a replacement for the inequalities that justify their use.

Accepted source and trust boundary. The corrected source manifest has SHA-256

`cbe4e29cf66b6a3a161e5ecac13b25c0c5465d32454966c1b2760c617844ae2b.`

A clean project rebuild completed on 19 October 2026 at 06:14:59 UTC. It recompiled all 556 selected local modules without reusing old project `.olean` objects. The all-declaration audit covered 77,040 constants, including 50,494 theorem constants, with axiom union `propext`, `Classical.choice`, `Quot.sound`. The final theorem source hash is

`6dbac3e70a6429287618d19c9fb180d403c9c7216d3d62b25b5d712fb0f85a79.`

The pinned Lean version is 4.24.0, commit `797c613eb9b6d4ec95db23e3e00af9ac6657f24b`; mathlib is pinned at `f897ebcf72cd16f89ab4577d0c826cd14afaafc7`. Pinned external dependency caches were reused and were not rebuilt from source in that run. The receipt therefore establishes a clean *project* rebuild, not a bootstrap of the compiler and all external dependencies. Standard library axioms and the trusted kernel/toolchain remain part of the trust base.

Source-provenance correction. An earlier incremental snapshot did not establish source/object correspondence: a later source edit was not represented by its previously built object, and the first isolated rebuild exposed the discrepancy. That snapshot is not the reproduction evidence for this manuscript. The source was corrected, a new immutable input manifest was selected, and the entire project build and audit were restarted successfully. The reproducibility documentation retains the superseded receipt and the corrected terminal receipt as distinct records.

What remains outside the claim. The finite polynomial trial is non-asymptotic because every residual term is retained and bounded on the entire interval. It is nevertheless organized around the decoupled dimer limit. This does not establish a new general nonperturbative phase theorem or a best-known coupling threshold. A broader literature comparison and external mathematical review remain appropriate. Further improvements could sharpen the finite certificate or the bootstrap, but are not needed for the theorem stated here. Reproducing the present result requires the pinned finite source checks, not an HPC spectrum calculation.

References

[1] OpenAI. The periodic spin-one Haldane gap. Pinned manuscript and source repository, 2026. Manuscript dated September 24, 2026. Proposition 3.1, Lemmas 4.1–4.2, Proposition 4.3, and Section 5.

[2] Simone Del Vecchio, Jürg Fröhlich, Alessandro Pizzo, and Stefano Rossi. Local iterative block-diagonalization of gapped Hamiltonians: a new tool in singular perturbation theory. *Journal of Mathematical Physics*, 63:073503, 2022. arXiv:2007.07667v2, Theorem 5.3.

[3] D. A. Yarotsky. Quasi-particles in weak perturbations of non-interacting quantum lattice systems. https://arxiv.org/abs/math-ph/0411042, 2004. arXiv:math-ph/0411042v1, Theorem 1.